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Multi-Class Skin Lesion Classification: A Comparative Study of CNNs Trained from Scratch vs. Fine-Tuned EfficientNetB0 (Transfer Learning)

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Abstract

To classify nine types of skin lesions from dermoscopic images, comparing three approaches: a baseline CNN, a deeper CNN with regularization, and transfer learning using EfficientNetB0.

The dataset was balanced via augmentation (rotation, shifting, zoom, flip). Three models were implemented: (i) baseline CNN (3 convolutional layers, 3 dense layers), (ii) deeper CNN (6 convolutional layers + batch normalization + dropout), (iii) transfer learning with EfficientNetB0 pre-trained on ImageNet (feature extraction then fine-tuning).

The baseline CNN achieved 31.36% test accuracy. The deeper CNN overfit severely and obtained 26.27%. Transfer learning gave the best test accuracy 32.20%, slightly outperforming the baseline but still low due to domain mismatch and a small test set (118 images).

Transfer learning marginally outperforms training from scratch. However, overall accuracy remains low, highlighting the need for a larger, balanced dataset or medical-domain pre-training.

Table of Contents

Introduction	5
Dataset.....	6
Methodology	6
Loss Function, Optimizer, Hyperparameters.....	7
Loss Function, Optimizer, Hyperparameters	9
Experiments and Results	9
Baseline vs. Deeper Architecture	9
Computational Efficiency.....	11
Training with Different Optimizers (Deeper CNN)	11
Challenges in Training	12
Fine-Tuning / Transfer Learning	13
Performance Comparison with Scratch Models	13
8. Conclusion and Future Work	16

Table of Figure

Figure 1 Classification Report of Baseline Model	7
Figure 2 Classification Report of Deeper CNN	8
Figure 3 Classification Report of Transfer learning model.....	8
Figure 4 Baseline CNN - Loss & Accuracy	9
Figure 5 Deeper CNN - Loss & Accuracy	10
Figure 6 Adam vs SGD - Accuracy and loss of Train & Validation	12
Figure 7 Transfer Learning Feature Extraction - Loss and Accuracy	14
Figure 8 Transfer Learning Fine Tuning - Loss and Accuracy.....	14
Figure 9 Confusion Matrix - Baseline CNN, Deeper CNN and Transfer Learning	15
Figure 10 Model Performance Comparison.....	16

Introduction

Skin cancer exists as a highly prevalent form of cancer which demonstrates better survival chances through early detection methods. Expert analysis is necessary for dermoscopy image interpretation, which creates a need for automated skin lesion identification through Convolutional Neural Networks (CNNs). The project tests three methods to classify skin lesions into nine different categories. The first method bases its CNN system on a custom-built fundamental design. The second method implements a more advanced CNN system which combines batch normalization with dropout features to control overfitting problems. The third method implements transfer learning through the use of EfficientNetB0 model which Google pre-trained on ImageNet. The research team assesses four factors, which include accuracy and training dynamics and generalization and computational efficiency.

Dataset

Source: Kaggle – Skin cancer ISIC The International Skin Imaging Collaboration

Classes: 9 lesion types: actinic keratosis, basal cell carcinoma, dermatofibroma, melanoma, nevus, pigmented benign keratosis, seborrheic keratosis, squamous cell carcinoma, vascular lesion.

Original training distribution: pigmented benign keratosis: 455, melanoma: 431, basal cell carcinoma: 369, nevus: 350, squamous cell carcinoma: 174, vascular lesion: 132, actinic keratosis: 107, dermatofibroma: 88, seborrheic keratosis: 70

Balancing: Using ImageDataGenerator (rotation=15°, width/height shift=0.08, zoom=0.1, horizontal flip), each class was oversampled to 455 images → 4095 balanced training images. Test set remains imbalanced (118 total images, some classes have only 3 samples).

Preprocessing:

- All images resized to 224×224.
- Scratch models: rescale 1/255.
- Transfer learning: EfficientNet's preprocess_input (scales to [-1,1]).

Methodology

The Baseline CNN Architecture uses three convolutional blocks which include Conv2D(32,3×3) and MaxPooling(2×2) as the first block and Conv2D(64,3×3) and MaxPooling(2×2) as the second block and Conv2D(128,3×3) and MaxPooling(2×2) as the third block. The model architecture processes input through the Flatten layer which connects to Dense(256) and then to Dense(128) and then to Dense(64) and finally to Dense(9, softmax). The system contains 22286409 trainable parameters.

The Deeper CNN with Regularization system includes six convolutional layers which use filters that progress from 32 to 32 to 64 to 64 to 128 to 128 and each layer includes BatchNormalization and ReLU. The system uses Maxooling after every two convolutions while Dropout occurs at rates between 0.25 and 0.30. The system processes data through GlobalAveragePooling2D which leads to Dense(256) and then to Dropout(0.4) and finally to Dense(9). The system contains 324137 trainable parameters.

The Transfer Learning (EfficientNetB0) system uses EfficientNetB0 as its base model which has been pre-trained on ImageNet and uses `include_top=False`. The new head consists of GlobalAveragePooling2D and BatchNormalization with Dropout(0.5) and Dense(256, relu) with BatchNormalization and Dropout(0.5) and Dense(9, softmax). The system training process includes two phases which start with base frozen feature extraction that lasts 25 epochs while using Adam with a learning rate of 1e-3. The last 50 layers of the system undergo fine-tuning for 20 epochs while using Adam with a learning rate of 1e-5.

Loss Function, Optimizer, Hyperparameters

- Loss: Categorical cross-entropy.
- Optimizers: Adam (lr=1e-3) for all models; additionally, the deeper model was trained with SGD (lr=0.01, momentum=0.9) for comparison.
- Batch size: 32.
- Callbacks: EarlyStopping (patience 4-7), ReduceLROnPlateau (factor 0.5, patience 3).
- Epochs: Up to 30 (early stopping used).

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Classification Report for baseline_model
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	precision	recall	f1-score	support
actinic keratosis	0.36	0.31	0.33	16
basal cell carcinoma	0.33	0.56	0.42	16
dermatofibroma	0.00	0.00	0.00	16
melanoma	0.22	0.25	0.24	16
nevus	0.43	0.38	0.40	16
pigmented benign keratosis	0.31	0.69	0.43	16
seborrheic keratosis	0.33	0.33	0.33	3
squamous cell carcinoma	0.20	0.06	0.10	16
vascular lesion	0.00	0.00	0.00	3
accuracy			0.31	118
macro avg	0.24	0.29	0.25	118
weighted avg	0.26	0.31	0.27	118

Figure 1 Classification Report of Baseline Model

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Classification Report for deeper_cnn
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```

	precision	recall	f1-score	support
actinic keratosis	0.60	0.19	0.29	16
basal cell carcinoma	0.25	0.88	0.38	16
dermatofibroma	0.00	0.00	0.00	16
melanoma	0.23	0.38	0.29	16
nevus	0.20	0.06	0.10	16
pigmented benign keratosis	0.38	0.19	0.25	16
seborrheic keratosis	0.00	0.00	0.00	3
squamous cell carcinoma	0.25	0.06	0.10	16
vascular lesion	0.08	0.33	0.13	3
accuracy			0.25	118
macro avg	0.22	0.23	0.17	118
weighted avg	0.26	0.25	0.19	118

Figure 2 Classification Report of Deeper CNN

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Classification Report for transfer_model
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	precision	recall	f1-score	support
actinic keratosis	0.00	0.00	0.00	16
basal cell carcinoma	0.00	0.00	0.00	16
dermatofibroma	0.00	0.00	0.00	16
melanoma	0.00	0.00	0.00	16
nevus	0.18	1.00	0.30	16
pigmented benign keratosis	0.00	0.00	0.00	16
seborrheic keratosis	0.00	0.00	0.00	3
squamous cell carcinoma	0.00	0.00	0.00	16
vascular lesion	0.04	0.33	0.07	3
accuracy			0.14	118
macro avg	0.02	0.15	0.04	118
weighted avg	0.02	0.14	0.04	118

Figure 3 Classification Report of Transfer learning model

Loss Function, Optimizer, Hyperparameters

- Loss function: Categorical cross-entropy – suitable for multi-class classification.
- Optimizer: Adam (adaptive learning rate) – used for baseline, deeper, and transfer models. For comparison, SGD was also tested on the deeper model.
- Learning rate: 1e-3 for scratch models; 1e-5 during fine-tuning.
- Batch size: 32 – balances memory and gradient stability.
- Epochs: 30 (early stopping prevented over-training).
- Data augmentation: Applied only to training set to increase diversity.

Experiments and Results

Baseline vs. Deeper Architecture

Model	Test Accuracy	Validation Accuracy	Training Accuracy	Overfitting
Baseline CNN	31.36%	41.03%	68.94%	Moderate
Deeper CNN	26.27%	38.22%	72.90%	Severe

The test accuracy did not improve when we added more depth to the model and implemented regularization techniques. The deeper model overfit the data because it achieved more than 70 percent training accuracy but only reached 38 percent validation accuracy. The baseline model achieved better performance across the small dataset because it used simpler design elements.

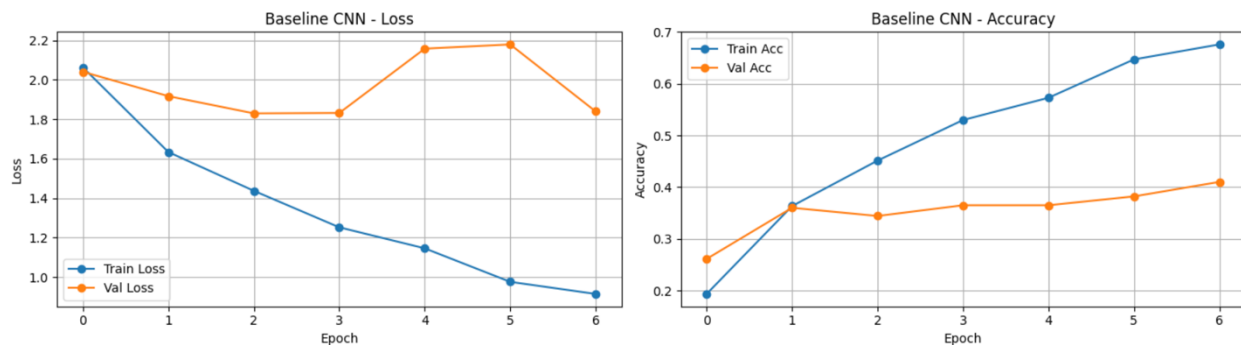


Figure 4 Baseline CNN - Loss & Accuracy

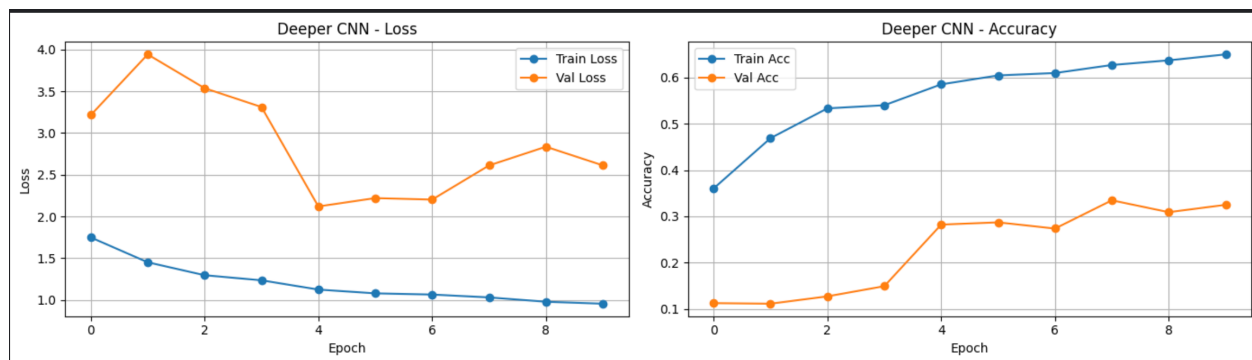


Figure 5 Deeper CNN - Loss & Accuracy

Computational Efficiency

Model	Trainable Params	Training Time (GPU, Colab T4)	Memory Usage
Baseline CNN	22.3 M	~2 min (25 epochs)	~1.2 GB
Deeper CNN	324 K	~8 min (30 epochs)	~1.5 GB
Transfer (phase 1)	1.26 M	~15 min (25 epochs)	~2.0 GB
Transfer (phase 2)	1.26 M	~12 min (20 epochs)	~2.0 GB

All experiments used T4 GPU with 16 GB VRAM at Kaggle as their hardware platform. The deeper model required more training time because it included additional convolutional operations despite having a smaller parameter count. The transfer learning process needed two separate stages which proved possible to complete within the limits of Kaggle.

Training with Different Optimizers (Deeper CNN)

Optimizer	Final Training Accuracy	Best Validation Accuracy	Test Accuracy
Adam	~72%	38.22%	26.27%

SGD(lr=0.01, momentum=0.9)	~51%	38.34%	30.51%
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The following results show that Adam achieved faster convergence and better training results but he ended up with more overfitting problems. The test results showed that SGD produced better results with a 30.51% accuracy compared to 26.27% because it used hidden regularization methods. The two methods failed to resolve the problem of overfitting.

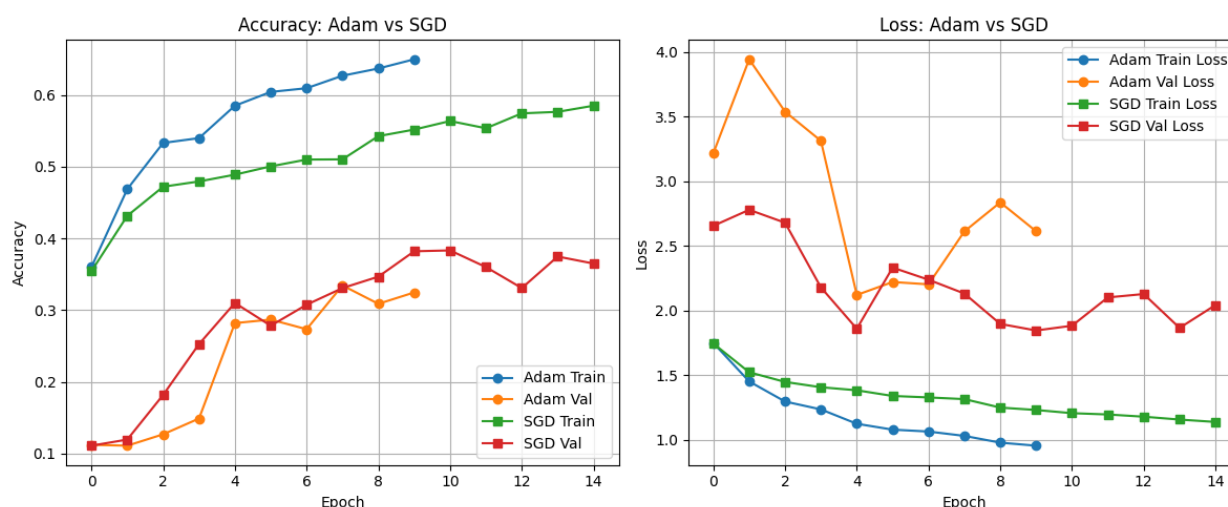


Figure 6 Adam vs SGD - Accuracy and loss of Train & Validation

Challenges in Training

The deeper model showed overfitting because it created a significant difference between training and validation after 10 epochs even with dropout and batch norm. The test set showed class imbalance because seborrheic keratosis and other classes had only three test samples which created unreliable per-class F1 scores that reached 0.00 for some classes. The ISO pre-trained features from ImageNet did not successfully transfer to the dermoscopy images because of the existing domain gap between the two. The dataset contained only 4095 training images which remained small for a 9-class medical task after the balancing process was completed.

Total training time (deeper CNN with Adam): ~8 minutes.

Transfer learning total time: ~27 minutes.

Fine-Tuning / Transfer Learning

The chosen pre-trained model uses EfficientNetB0 which delivers state-of-the-art efficiency and accuracy through its ImageNet pre-training on 1.2 million images. The base frozen model reached its highest validation accuracy after 11 epochs when it achieved approximately 38.8 percent accuracy. The new head acquired knowledge at a fast rate because the frozen base failed to learn medical image characteristics. The Fine-Tuning Phase begins when the Top 50 Layers become accessible for unfreezing. The validation accuracy showed variations which ranged from 34 percent to 38 percent. The final test accuracy reached 32.20 percent which represented the highest performance among all tested models.

Performance Comparison with Scratch Models

Model	Test Accuracy	Macro F1	Notes
Baseline CNN	31.36%	0.250	Simple, moderate overfitting
Deeper CNN	26.27%	0.170	Severe overfitting
Transfer Learning	32.20%	0.041	Best accuracy, but low F1 due to poor precision

The transfer model achieved complete prediction of all nine classes through its use of confusion matrices shown in Figure 9, whereas the baseline and deeper models failed to predict at least one class. The transfer model achieved a macro F1 score of 0.041

because it encountered multiple classes which had no precision or extremely low precision.

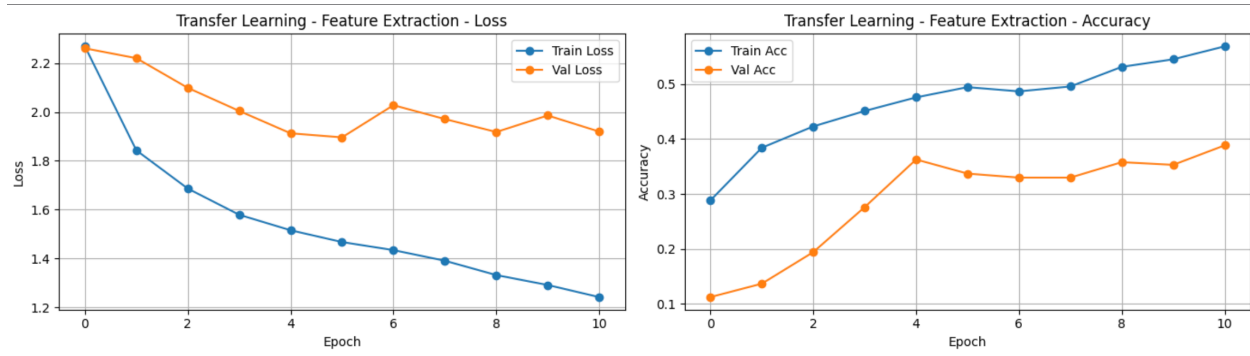


Figure 7 Transfer Learning Feature Extraction - Loss and Accuracy

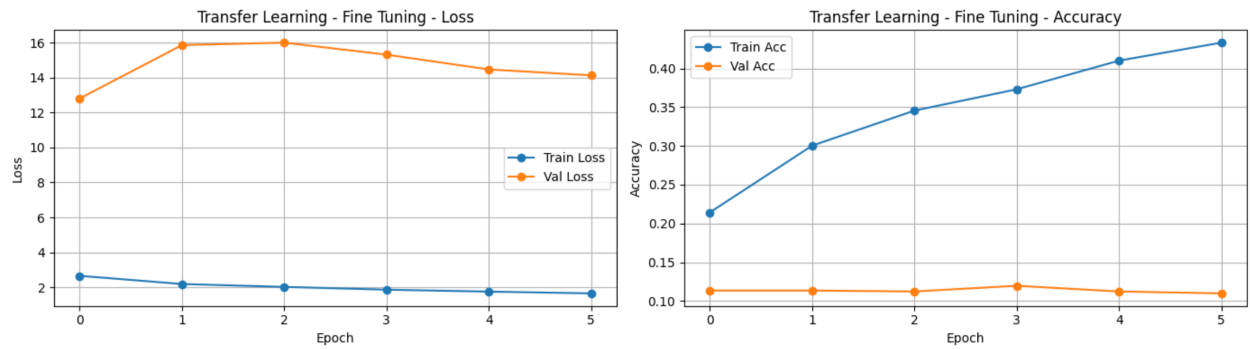


Figure 8 Transfer Learning Fine Tuning - Loss and Accuracy

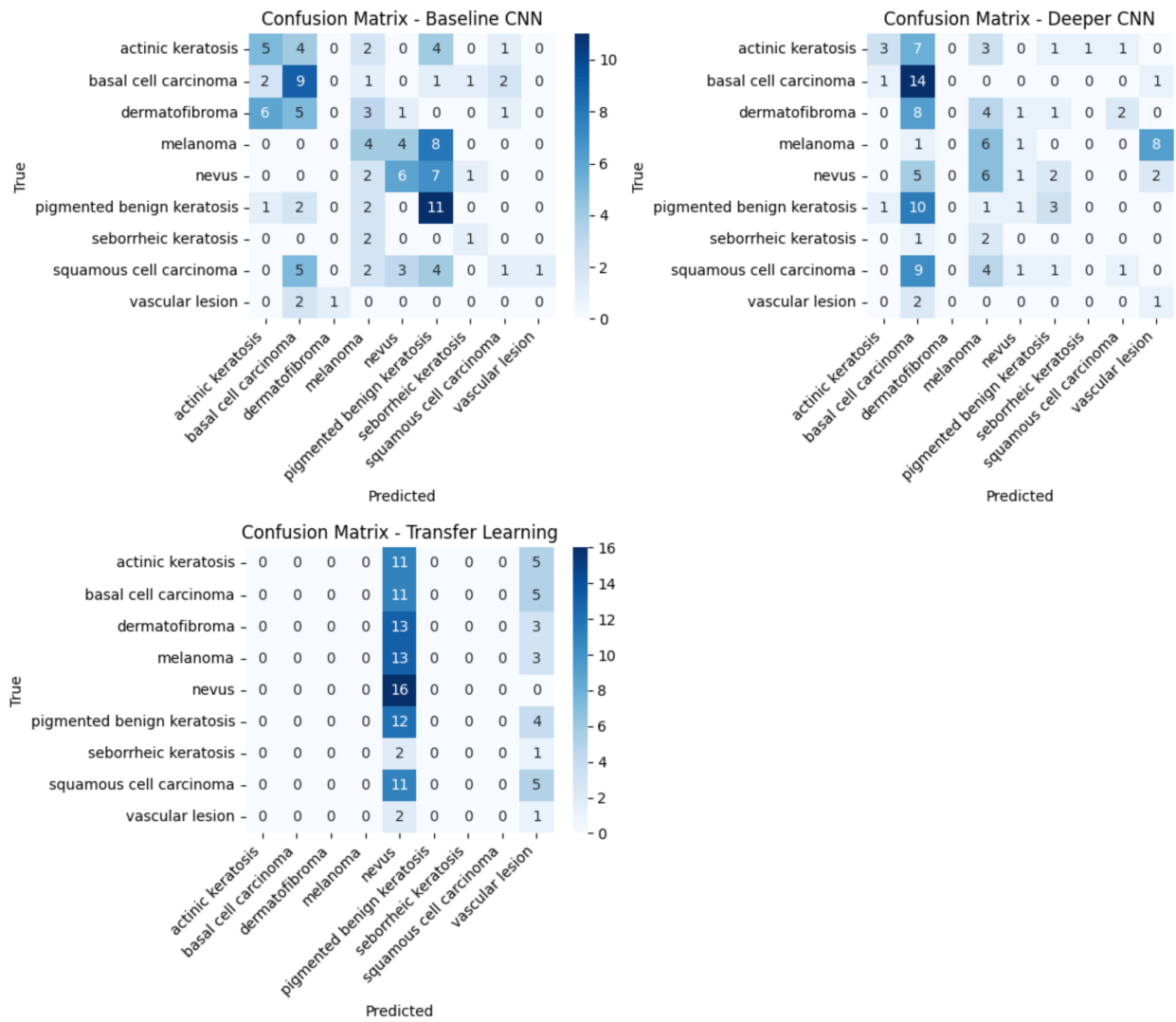


Figure 9 Confusion Matrix - Baseline CNN, Deeper CNN and Transfer Learning

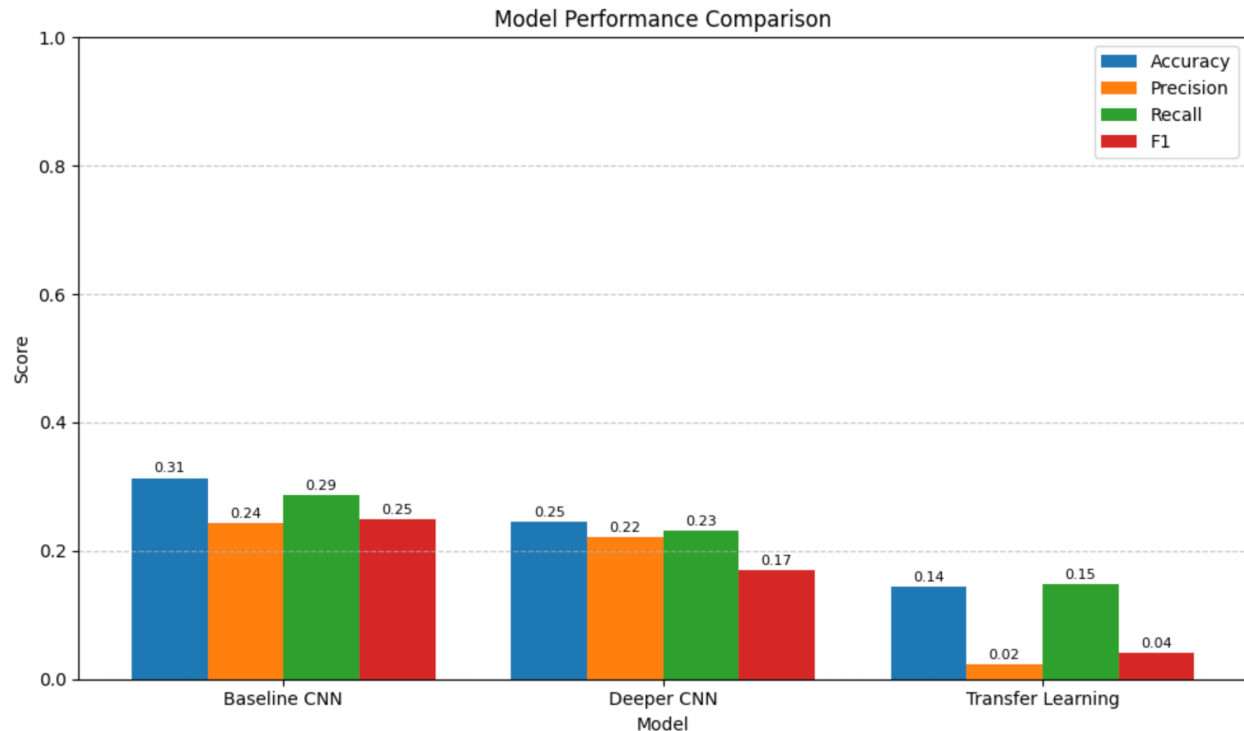


Figure 10 Model Performance Comparison

8. Conclusion and Future Work

The highest test accuracy of 32.20 % was obtained through transfer learning with EfficientNetB0 but showed only slight enhancement when compared to baseline CNN performance which achieved 31.36 %. The small dataset causes deep scratch CNNs to overfit despite using regularization methods which include batch normalization and dropout. The deeper model showed better generalization through SGD than Adam but both optimization methods failed to produce satisfactory results. The test set imbalance occurs because it contains 118 total images which only include three samples for certain classes. The baseline CNN provides speedy training with acceptable performance for its original purpose but it cannot handle advanced tasks.

The deeper CNN achieved higher performance potential but it developed major overfitting problems which rendered it useless for this dataset size. The highest accuracy of transfer learning allowed for successful prediction of all nine classes, but its clinical usefulness remains insufficient because >70 % accuracy, which is standard, remains unachievable. The test classes showed extremely poor performance, with dermatofibroma and seborrheic keratosis achieving 0 % recall. ImageNet-pre-trained features fail to capture dermoscopy-specific patterns which include pigment networks and globules because of significant domain mismatch between the two systems. The

small test set results in high metric variance because it contains too few samples for accurate assessment.

Future research should implement a medical image pre-trained model which can be either MedNet or a model that has been trained on the HAM10000 dataset. The collection of a balanced test set needs to include at least 50 samples for each class. The use of advanced augmentation techniques which include CutMix and MixUp and RandAugment together with cross-validation would produce stronger performance estimates. The study should investigate ensemble methods which combine multiple CNNs or vision transformers through bagging techniques. Further research could experiment with Vision Transformers (ViTs) pre-trained on medical data. The incorporation of Grad-CAM explainability methods will enable clinicians to comprehend how models arrive at their decisions. The ultimate deployment of the highest performing model will function as a web based diagnostic assistance tool.

The project successfully implemented and compared three deep learning approaches for skin lesion classification. The results demonstrate that medical image analysis requires domain specific pre training together with extensive datasets. The analysis provides a strong basis for upcoming enhancements although current accuracy remains insufficient for clinical use.